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Inventor(s)

KIM; Taehwi et al.

ORGANIC LIGHT EMITTING DISPLAY DEVICE

Abstract

Provided is an organic light emitting display device. The organic light emitting display device includes a substrate including an active area and an inactive area at least partially surrounding the active area, an organic light emitting element in the active area, and a conductive line disposed to have a vertically uneven shape in the inactive area and provided to transmit power to the organic light emitting element. The conductive line covers an area including at least one or more grooves in which an inorganic layer is dug.

Inventors: KIM; Taehwi (Busan, KR), HONG; HyunSeok (Gumi-si, KR), KANG; MinJi (Seocheon-gun, KR)

Applicant: LG Display Co., LTD. (Seoul, KR)

Family ID: 76438407

Assignee: LG Display Co., LTD. (Seoul, KR)

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Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to an organic light emitting display device.

Description of the Related Art

[0002] Image display devices that implement various types of information on a screen are regarded as a core technology of the information communication era. Such image display devices are becoming thinner, lighter, and more portable, while having higher performance characteristics.

[0003] An organic light emitting element is a self-light emitting element using a thin light emitting layer between electrodes and is advantageous in terms of thinning thereof. A general organic light emitting display device has a structure in which a pixel driving circuit and an organic light emitting element are formed on a substrate, and light emitted from the organic light emitting element passes through the substrate or a barrier layer to thereby display an image.

[0004] Since the organic light emitting display device is implemented without a separate light source device, it can be easily implemented as a flexible display device. In this case, a flexible material such as plastic or metal foil may be used as the substrate of the organic light emitting display device.

BRIEF SUMMARY

[0005] The inventors of the present disclosure have identified one or more problems in the related art. Namely, the inventors recognized that because performance of the organic light emitting element degrades when the element is exposed to moisture, a moisture-proof design of an outer portion of the display device is beneficial. Accordingly, the inventors have provided a display device that implements various moisture-proof designs applied to, for example, the outer portion of the display device. The inventors have also provided a display device having a reduction in width at the outer portion (e.g., bezel) of the display device to meet the industries and the consumer desires for an aesthetically pleasing device.

[0006] Taking into consideration the problems faced in the related art (not limited to those described above), the inventors of the present disclosure provides in one or more embodiments, an organic light emitting display device having a reduced bezel structure.

[0007] Technical benefits of the present disclosure are not limited to the above-mentioned benefits, and other benefits, which are not mentioned above, can be clearly understood by those skilled in the art from the following descriptions.

[0008] According to an aspect of the present disclosure, an organic light emitting display device includes a substrate including an active area and an inactive area adjacent to the active area, an organic light emitting element in the active area, and a conductive line disposed to have a vertically uneven shape in the inactive area and provided to transmit power to the organic light emitting element. The conductive line covers an area including at least one or more grooves in which an inorganic layer is dug.

[0009] The inorganic layer may include at least one or more of a buffer layer, a gate insulating layer, an interlayer insulating layer, and a protective layer. In this case, the buffer layer may include a multi-buffer and an active buffer. The active buffer may be removed from the grooves, and the multi-buffer is not removed from the grooves.

[0010] The organic light emitting display device may further include an encapsulation layer covering the organic light emitting element. The encapsulation layer may include a first inorganic layer on the organic light emitting element, an organic layer on the first inorganic layer, and a second inorganic layer on the organic layer.

[0011] The organic light emitting display device may further include at least two or more blocking structures provided to block a flow of the organic layer of the encapsulation layer, in the inactive area.

[0012] The conductive line may not vertically overlap with a blocking structure disposed at an outermost position among the blocking structures.

[0013] The blocking structures may include a first blocking structure, a second blocking structure which is farther from the active area than the first blocking structure, and a third blocking structure which is farther from the active area than the second blocking structure. The conductive line may not vertically overlap with the third blocking structure.

[0014] The first blocking structure may be formed of a same or substantially the same material as a planarization layer and a bank of the active area. The second blocking structure may be formed by sequentially stacking materials the same as those of a planarization layer, a bank, and a spacer of the active area. The third blocking structure may be formed of a same or substantially the same material as a planarization layer of the active area.

[0015] Other detailed matters of the embodiments are included in the detailed description and the drawings.

[0016] According to embodiments of the present disclosure, an organic light emitting display device in which damage to a conductive line of an outer portion is improved can be provided. More specifically, according to embodiments of the present disclosure, a structure capable of preventing damage to conductive lines and the resultant driving failure, while having a narrow bezel, can be provided. Accordingly, moisture-proof performance and long-term reliability of the organic light emitting display device according to the embodiment of the present disclosure can be improved.

[0017] The effects according to the present disclosure are not limited to the contents exemplified above, and more various effects are included in the present specification.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The above and other aspects, features and other advantages of the present disclosure will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0019] FIG. 1 is a plan view illustrating an organic light emitting display device according to an embodiment of the present disclosure;

[0020] FIG. 2 is a cross-sectional view illustrating a portion of an active area of the organic light emitting display device according to an embodiment of the present disclosure;

[0021] FIG. 3 is a view illustrating a portion of an inactive area of an organic light emitting display device according to another embodiment of the present disclosure;

[0022] FIG. 4 is a view illustrating an outer portion structure applied to the organic light emitting display device according to another embodiment of the present disclosure; and

[0023] FIG. 5 is an enlarged view of portion X of FIG. 4.

DETAILED DESCRIPTION

[0024] Advantages and characteristics of the present disclosure and a method of achieving the advantages and characteristics will be clear by referring to embodiments described below in detail together with the accompanying drawings. However, the present disclosure is not limited to the embodiments disclosed herein but will be implemented in various forms. The embodiments are provided by way of example only so that those skilled in the art can fully understand the disclosures of the present disclosure and the scope of the present disclosure.

[0025] The shapes, sizes, ratios, angles, numbers, and the like illustrated in the accompanying drawings for describing the embodiments of the present disclosure are merely examples, and the present disclosure is not limited thereto. Like reference numerals generally denote like elements throughout the specification. Further, in the following description of the present disclosure, a detailed explanation of known related technologies may be omitted to avoid unnecessarily obscuring the subject matter of the present disclosure. The terms such as “including,” “having” used herein are generally intended to allow other components to be added unless the terms are used with the term “only.” Any references to singular may include plural unless expressly stated otherwise. Components are interpreted to include an ordinary error range even if not expressly stated.

[0026] When the position relation between two parts is described using the terms such as “on,” “above,” “below,” and “next,” one or more parts may be positioned between the two parts unless the terms are used with the term “immediately” or “directly.” When an element or layer is disposed “on” another element or layer, another layer or another element may be interposed directly on the other element or therebetween.

[0027] When it is described that any component is “connected” or “coupled” to another component, the component may be directly connected or coupled to another component, but it should be understood that any other component may also be “interposed” therebetween, or the respective components may be “connected,” “coupled” or “joined” to each other through a third component.

[0028] Although the terms “first,” “second,” and the like are used for describing various components, these components are not confined by these terms. These terms are merely used for distinguishing one component from the other components. Therefore, a first component to be mentioned below may be a second component in a technical concept of the present disclosure.

[0029] Like reference numerals generally denote like elements throughout the specification.

[0030] A size and a thickness of each component illustrated in the drawing are illustrated for convenience of description, and the present disclosure is not limited to the size and the thickness of the component illustrated. The features of various embodiments of the present disclosure can be partially or entirely adhered to or combined with each other and can be interlocked and operated in technically various ways, and the embodiments can be carried out independently of or in association with each other.

[0031] Hereinafter, a display device according to embodiments of the present disclosure will be described in detail with reference to accompanying drawings.

[0032] FIG. 1 is a plan view illustrating an organic light emitting display device according to an embodiment of the present disclosure.

[0033] With reference to FIG. 1, an organic light emitting display device **100** includes at least one active area A/A, and an array of pixels is disposed in the active area. One or more inactive areas I/A may be disposed around the active area. That is, the inactive area may be adjacent to one or more side surfaces of the active area. In FIG. 1, the inactive area at least partially surrounds the active area having a rectangular shape. However, the shape of the active area, and a shape and disposition of the inactive area adjacent to the active area are not limited to an example shown in FIG. 1. The active area and the inactive area may be in a form suitable for a design of an electronic device on which the display device **100** is mounted. The shape of the active area is, for example, pentagonal, hexagonal, circular, or elliptical.

[0034] Each pixel in the active area A/A may be associated with a pixel circuit. The pixel circuit may include one or more switching transistors and one or more driving transistors. Each pixel circuit may be electrically connected to signal lines (a gate line, a data line and the like) to communicate with a gate driver, a data driver and the like located in the inactive area.

[0035] The gate driver and the data driver may be implemented as a thin film transistor (TFT) in the inactive area I/A. Such a driver may be referred to as a gate-in-panel (GIP). In addition, some components, such as data driver ICs, are mounted on a separate printed circuit board, and may be combined with a connection interface (a pad, a bump, a pin or the like) disposed in the inactive area through circuit films such as a flexible printed circuit board (FPCB), a chip-on-film (COF), a tape-carrier-package (TCP), and the like. The printed circuit (COF, PCB or the like) may be located behind the display device **100**.

[0036] The organic light emitting display device **100** may include various additional elements for generating various signals or driving pixels in the active area. The additional elements for driving the pixels may include an inverter circuit, a multiplexer, an electrostatic discharge (ESD) circuit, and the like. The organic light emitting display device **100** may also include additional elements associated with functions other than pixel driving. For example, the organic light emitting display device **100** may include additional elements for providing a touch sensing function, a user authentication function (e.g., fingerprint recognition), a multi-level pressure sensing function, a tactile feedback function, and the like. The additional elements described above may be located in the inactive area and/or an external circuit connected to the connection interface.

[0037] The organic light emitting display device according to an embodiment of the present disclosure may include a substrate **101** on which a thin film transistor and an organic light emitting element are arranged, an encapsulation layer **120**, a polarization layer **145**, and the like.

[0038] The substrate **101** supports various components of the organic light emitting display device **100**. The substrate **101** may be formed of a transparent insulating material, for example, an insulating material such as glass, plastic or the like. The substrate (array substrate) is also referred to as a concept including an element and a functional layer formed thereon, for example, a switching TFT, a driving TFT connected to the switching TFT, an organic light emitting element connected to the driving TFT, a protective film, and the like.

[0039] The organic light emitting element is disposed on the substrate **101**. The organic light emitting element includes an anode, an organic light emitting layer, and a cathode. The organic light emitting element may be configured as a single light emitting layer structure that emits one light, or may be configured as a structure composed of a plurality of light emitting layers to emit white light. When the organic light emitting element emits white light, a color filter may be further provided. The organic light emitting element may be formed in a central portion of the substrate **101** to correspond to the active area.

[0040] The encapsulation layer **120** may cover the organic light emitting element. The encapsulation layer protects the organic light emitting element from external moisture or oxygen.

[0041] The organic light emitting display device **100** may be configured to have a plurality of pixels, and each of the plurality of pixels may include a plurality of sub-pixels. In this case, the sub-pixel is a minimum unit for expressing one color.

[0042] One sub-pixel circuit may include a plurality of transistors, a capacitor, and a plurality of lines. The sub-pixel circuit may be configured of two transistors and one capacitor (2T1C), but it is not limited thereto and may be implemented as a sub-pixel circuit configured of 4T1C, 7T1C, 6T2C, or the like. In addition, the sub-pixel may be implemented to be suitable for the organic light emitting display device **100** of a top emission type.

[0043] FIG. 2 is a cross-sectional view illustrating a portion of an active area of the organic light emitting display device according to an embodiment of the present disclosure.

[0044] The organic light emitting display device **100** may include the substrate **101**, a thin film transistor, an organic light emitting element, and various functional layers.

[0045] The substrate **101** serves to support and protect components of the organic light emitting display device **100** disposed thereon. The substrate **101** may be a flexible substrate formed of a ductile material having flexible characteristics. The substrate **101** may be a glass or plastic substrate. In the case of the plastic substrate, a polyimide-based or polycarbonate-based material may be used to have flexibility. In particular, polyimide can be applied to high temperature processes and is widely used for the plastic substrate because it is a material that can be coated.

[0046] A buffer layer **103** is a functional layer for protecting the electrode/line from impurities such as alkali ions or the like flowing out from the substrate **101** or lower layers. The buffer layer **103** may be formed of silicon oxide (SiOx), silicon nitride (SiNx), or a multilayer thereof. The buffer layer **103** may include a multi-buffer and/or an active buffer. The multi-buffer may be formed by alternately stacking silicon nitride (SiNx) and silicon oxide (SiOx), and may delay diffusion of moisture and/or oxygen impregnated into the substrate **101**. The active buffer protects a semiconductor layer **102** of the transistor and functions to block various kinds of defects introduced from the substrate **101**. The active buffer may be formed of amorphous silicon (a-Si) or the like.

[0047] The thin film transistor includes a gate electrode **104**, source and drain electrodes **108**, and the semiconductor layer **102**. The semiconductor layer **102** may be formed of amorphous silicon or polycrystalline silicon. Polycrystalline silicon has superior mobility than amorphous silicon, and thus, has low energy consumption and excellent reliability. Recently, oxide semiconductors have been spotlighted for their excellent mobility and uniformity. The semiconductor layer **102** may include a source region including a p-type or n-type impurity, a drain region, and a channel between the source region and the drain region, and may include a low concentration-doped region between the source region and the drain region adjacent to the channel.

[0048] A gate insulating layer **105** is an insulating film composed of a single layer or multiple layers of silicon oxide (SiOx) or silicon nitride (SiNx), and is provided so that a current flowing through the semiconductor layer **102** does not flow to the gate electrode **104**.

[0049] The gate electrode **104** serves as a switch to turn-on or turn-off the thin film transistor based on an electrical signal transmitted from the outside through the gate line, and may be composed of a single layer or multiple layers of copper (Cu), aluminum (Al), molybdenum (Mo), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd) or the like, which is a conductive metal, or alloys thereof. The source and drain electrodes **108** are connected to the data lines and allow electrical signals transmitted from the outside to be transferred from the thin film transistor to the organic light emitting element. The source and drain electrodes **108** may be composed of a single layer or multiple layers of copper (Cu), aluminum (Al), molybdenum (Mo), chromium (Cr), gold (Au), titanium (Ti), nickel (Ni), neodymium (Nd) or the like, which is a metallic material, or alloys thereof.

[0050] To insulate the gate electrode **104** and the source and drain electrodes **108** from each other, an interlayer insulating layer **106** composed of a single layer or multiple layers of silicon oxide (SiOx) or silicon nitride (SiNx) may be disposed between the source and drain electrodes **108**.

[0051] A protective layer **107** formed of an inorganic insulating film such as silicon oxide (SiOx) or silicon nitride (SiNx) may be positioned on the thin film transistor. The protective layer **107** may prevent unnecessary electrical connections between components of the thin film transistor and prevent contamination or damage from the outside. The protective layer **107** may be omitted depending on a configuration and characteristics of the thin film transistor and the organic light emitting element.

[0052] For convenience of description, only a driving thin film transistor is illustrated among various thin film transistors, but a switching thin film transistor, a capacitor, and the like may also be included in the active area. When a signal is applied from a gate line to the switching thin film transistor, the switching thin film transistor transmits the signal from a data line to a gate electrode of the driving thin film transistor. The driving thin film transistor transmits a current transmitted through a power supply line to an anode by the signal received from the switching thin film

transistor, and light emission is controlled by the current transmitted to the anode.

[0053] A planarization layer **109** is disposed on the thin film transistor. The planarization layer **109** protects the thin film transistor, alleviates a step caused by the thin film transistor, and reduces parasitic-capacitance generated between the thin film transistor, the gate line and the data line, and the organic light emitting element. The planarization layer **109** may be formed of one or more of acrylic resin, epoxy resin, phenolic resin, polyamides resin, polyimides resin, unsaturated polyesters resin, polyphenylene resin, polyphenylene sulfides resin, and benzocyclobutene.

[0054] The organic light emitting element is disposed on the planarization layer **109**. The organic light emitting element includes an anode **112**, a light emitting unit **114**, and a cathode **116**. The anode **112** may be disposed directly on the planarization layer **109**. The anode **112** is an electrode that serves to supply holes to the light emitting unit **114** and may be electrically connected to the thin film transistor through a contact hole in the planarization layer **109**. The anode **112** may be formed of indium tin oxide (ITO), indium zinc oxide (IZO), or the like, which is a transparent conductive material. When the organic light emitting display device **100** is a top emission type in which it emits light upwardly, it may further include a reflective layer such that the emitted light is more smoothly emitted in a direction toward an upper portion thereof where the cathode **133** is disposed. The anode **112** may be a two-layer structure in which a transparent conductive layer formed of a transparent conductive material and a reflective layer are sequentially stacked, or a three-layer structure in which a transparent conductive layer, a reflective layer, and a transparent conductive layer are sequentially stacked. The reflective layer may be formed of silver (Ag) or an alloy including silver.

[0055] In some embodiments, a bank **110** is disposed on the anode **112** and the planarization layer **109** and may define an area that emits light. The bank **110** is formed by photolithography after forming a photoresist on the anode **112**. Photoresist refers to a photosensitive resin whose solubility in a developer is changed by an action of light, and a specific pattern can be obtained by exposing and developing the photoresist.

[0056] A fine metal mask (FMM) which is a deposition mask, may be used to form the light emitting unit **114** of the organic light emitting element. At this time, to prevent damage that may occur due to contact with the deposition mask disposed on the bank **110** and to maintain a constant distance between the bank **110** and the deposition mask, a spacer **111** formed of one of polyimide which is a transparent organic material, photo acryl, and benzocyclobutene (BCB) may be disposed on the bank **110**.

[0057] The light emitting unit **114** is disposed between the anode **112** and the cathode **116**. The light emitting unit **114** serves to emit light and may include at least one of a hole injection layer (HIL), a hole transport layer (HTL), a light emitting layer (EML), an electron transport layer (ETL), and an electron injection layer (EIL), and some components of the light emitting unit **114** may be omitted depending on a structure or characteristics of the organic light emitting display device **100**.

[0058] The cathode **116** is disposed on the light emitting unit **114** and serves to supply electrons to the light emitting unit **114**. Since the cathode **116** needs to supply electrons, it may be formed of a metallic material such as magnesium (Mg), silver-magnesium (Ag:Mg), which is a conductive material having a low work function. In a case where the organic light emitting display device **100** is a top emission type, the cathode **116** may be a transparent conductive oxide such as indium tin oxide (ITO), indium zinc oxide (IZO), indium tin zinc oxide (ITZO), zinc oxide (ZnO), and tin oxide (TO).

[0059] The encapsulation layer **120** for preventing oxidation or damage due to moisture, oxygen, or impurities introduced from the outside may be disposed on the organic light emitting element. The encapsulation layer **120** may be formed by stacking a plurality of inorganic layers, a foreign material compensation layer, and a plurality of barrier films. The inorganic layers may be disposed on the entire surface of an upper portion of the organic light emitting element, and may be formed

of one of silicon nitride (SiNx) and aluminum oxide (Al₂O₃) which is an inorganic material. An additional inorganic layer may be stacked and disposed on the foreign material compensation layer. The foreign material compensation layer is disposed on the inorganic layer, and an organic material such as silicon oxycarbon (SiOCz), acrylic, or epoxy-based resin may be used for the foreign material compensation layer. When a defect occurs due to a crack generated by a foreign material or particles that may be generated during a process, it may compensate for those by covering a crack and a foreign material.

[0060] The barrier film may be disposed on the encapsulation layer **120** to further delay penetration of oxygen and moisture from the outside. The barrier film is configured in the form of a light-transmissive and double-sided adhesive film, and may be composed of any one of olefin-based, acrylic-based, and silicon-based insulating materials. Alternatively, a barrier film composed of any one of COP (cycloolefin polymer), COC (cycloolefin copolymer) and PC (Polycarbonate) may be further used. A touch electrode (a panel), a polarizing film, a cover glass, and the like may be further positioned on the encapsulation layer **120**.

[0061] FIG. **3** is a view illustrating a portion of an inactive area of an organic light emitting display device according to another embodiment of the present disclosure.

[0062] The inactive area I/A may be located outside the active area A/A as illustrated in FIG. **1**, and a driving circuit (e.g., GIP), a power supply line, and the like may be disposed thereon. Although the pixel circuit and the light emitting element are not disposed in the inactive area I/A, the substrate **101** and organic/inorganic functional layers **103**, **105**, **107**, **109**, **120**, and the like may exist therein. In addition, materials used in configuring the active area A/A may be disposed in the inactive area I/A for other uses. For example, a metal **104'** the same as the gate electrode of the TFT of the active area, or a metal **108'** the same as the source/drain electrode may be disposed in the inactive area I/A for lines or electrodes. Furthermore, a metal **112'** the same as one electrode (e.g., an anode) of an organic light emitting diode may be disposed in the inactive area I/A for lines and electrodes.

[0063] The encapsulation layer **120** covers an upper portion of the organic light emitting element. The encapsulation layer may be composed of an inorganic layer formed of glass, metal, an aluminum oxide (AlOx) or silicon (Si)-based material, or may have a structure in which an organic layer and an inorganic layer are alternately stacked. Inorganic layers **121-1** and **121-2** serve to block penetration of moisture and oxygen, and an organic layer **122** serves to flatten surfaces of the inorganic layers **121-1** and **121-2**.

[0064] The organic layer **122** has a certain degree of flowability, and may flow to the outside of the inactive area during application. Accordingly, blocking structures **190-1**, **190-2** and **190-3** are disposed to control spreading of the organic layer **122** to the inactive area I/A. FIG. **3** illustrates that three blocking structures **190-1**, **190-2** and **190-3**, that is, three dams **190-1**, **190-2** and **190-3** are disposed, but the number of dams may be varied. Further, in some embodiments, the dams **190-1**, **190-2**, and **190-3** may be disposed to surround the active area A/A or may be disposed within the active area A/A. The dams **190-1**, **190-2** and **190-3** may be formed of multiple layers using at least one or more materials. For example, the dams **190-1**, **190-2** and **190-3** can be manufactured using materials used in forming the planarization layers **109**, the bank **110**, and the spacer.

[0065] On the encapsulation layer **120**, a touch sensing line **154** and a touch driving line **152** may be alternately disposed with a touch insulating layer **158** interposed therebetween. Mutual capacitance (C_m) is formed at an overlapping location of the touch sensing line **154** and the touch driving line **152**. Accordingly, the mutual capacitance C_m serves as a touch sensor by charging charges by a touch driving pulse supplied to the touch driving line **152** and discharging the charged charges to the touch sensing line **154**.

[0066] The touch driving line **152** includes a plurality of first touch electrodes and first bridges that electrically connect the first touch electrodes to one another. The plurality of first touch electrodes are spaced apart at regular intervals along a first direction. Each of the plurality of first touch

electrodes is electrically connected to the first touch electrode adjacent thereto through a first bridge. The first bridge is disposed on the same layer as the first touch electrode and can be electrically connected to the first touch electrode without a separate contact hole.

[0067] The touch sensing line **154** includes a plurality of second touch electrodes and second bridges that electrically connect the plurality of second touch electrodes. The plurality of second touch electrodes are spaced apart at regular intervals along a second direction. Each of the plurality of second touch electrodes is electrically connected to the second touch electrode adjacent thereto through a second bridge. The second bridge is disposed on a layer different from the second touch electrode and can be electrically connected to the second touch electrode through a contact hole.

[0068] Each of the first and second touch electrodes and the first and second bridges may be formed in a single layer or multilayer structure with the use of a conductive layer having high corrosion resistance and acid resistance and good conductivity, such as aluminum (Al), titanium (Ti), copper (Cu), or molybdenum (Mo). For example, each of the first and second touch electrodes and the first and second bridges is formed as a stacked three-layer structure such as Ti/Al/Ti or Mo/Al/Mo.

[0069] Each of the touch driving line **152** and the touch sensing line **154** is connected to a touch driver through a routing line and a touch pad disposed in the inactive area. Accordingly, the routing line transmits the touch driving pulse generated by the touch driver to the touch driving line **152** through the touch pad, and transmits a touch signal from the touch sensing line **154** to the touch pad.

[0070] Various circuits and electrodes/lines disposed in the inactive area I/A may be formed of the gate metal **104'** and/or the source/drain metal **108'**. At this time, the gate metal **104'** and the gate electrode of the TFT are formed using the same or substantially the same material in the same process, and the source/drain metal **108'** and the source/drain electrode of the TFT are formed using the same or substantially the same material in the same process.

[0071] For example, the source/drain metal may be used as a power supply (e.g., low level power supply (VSS), high level power supply (VDD) or the like) line **108'**. At this time, the power supply line **108'** is connected to the metal layer **112'**, and the cathode **116** of the organic light emitting element is supplied with power through connection with the source/drain metal **108'** and the metal layer **112'**. The metal layer **112'** may come into contact with the power supply line **108'**, and extend along a sidewall of an outermost portion of the planarization layer **109** to thereby be in contact with the cathode **116** on the planarization layer **109**. The metal layer **112'** and the anode **112** of the organic light emitting diode may be formed using the same or substantially the same material in the same process.

[0072] The inventors have discovered vulnerable points in an outer portion structure of the organic light emitting display device as shown in FIG. 3. One of the vulnerable points is moisture permeation which is shown in the vicinity (W) of an end portion of the power supply line **108'**.

[0073] The power supply line **108'** is designed to be as wide as possible in order to reduce resistance and stably supply power. Accordingly, the end portion of the power supply line **108'** overlaps with the blocking structure **190-3** disposed at an outermost position to a predetermined degree. Since adhesion between the power supply line **108'** and the protective layer **107** disposed thereon is not high, an end of the insulating layer **158** disposed outwardly of the blocking structure **190-3** is contracted, thereby lifting the inorganic layers **121-1** and **121-2** disposed below.

Consequently, a void is caused between the power supply line **108'** and the protective layer **107** disposed thereon. In addition, the inorganic layers **121-1** and **121-2** may not completely cover the power supply line **108'** due to process variations and shadows of the inorganic layers **121-1** and **121-2** of the encapsulation layer.

[0074] When moisture penetrates into the void and/or incompletely covered portions caused for these reasons, corrosion of the power supply line **108'** occurs. As a result, the line metal may be eluted or deformed, and interlayer delamination may also diffuse, thereby leading to poor driving

of the display device.

[0075] The inventors of the present disclosure have recognized vulnerabilities and dangers described above, and have invented a structure to prevent moisture permeation and corrosion of conductive lines in the outer portion.

[0076] FIG. 4 is a view illustrating an outer portion structure applied to the organic light emitting display device according to another embodiment of the present disclosure. FIG. 5 is an enlarged view of portion X of FIG. 4.

[0077] Among components in FIG. 4, a duplicate description of the same components as those in FIGS. 1 to 3 will be omitted. The organic light emitting display device according to the embodiment may include a structure, in which a line width is adjusted so that a conductive line **408'** of the inactive area is not placed at an outer edge and an increase in resistance due to the conductive line is suppressed. A structure of the conductive line may be provided by imparting a shape in which the conductive line **408'** has unevenness in a vertical direction. As an example, the conductive line **408'** in the outer portion is provided only up to an inside of the blocking structure disposed at an outermost position, so that it has a surface area maintained the same as or similarly to the previous one, while having unevenness using a groove (a recessed area) from which the lower inorganic layers are removed.

[0078] The organic light emitting display device according to the embodiment includes a substrate **401** having an active area A/A and an inactive area I/A at least partially surrounding the active area A/A. The organic light emitting display device includes an organic light emitting element in the active area A/A. The organic light emitting display device further includes the conductive line **408'** disposed to have a vertically uneven shape in the inactive area I/A and configured to transmit power to the organic light emitting element. As shown in FIGS. 4 and 5, the conductive line **408'** has an uneven shape in a vertical direction. In some embodiments, at least some portions of the conductive line **408'** are not deposited on the same layer. That is, some portions of the conductive line **408'** may not be coplanar with each other. In other embodiments, at least some portions of the conductive line **408'** may extend through to contact other layers than that the conductive line **408'** is deposited on. For example, some portions of the conductive line **408'** may extend to contact the substrate **401**.

[0079] The conductive line **408'** may be a power supply line that transmits a low level power supply (V.sub.SS) to the organic light emitting element. The conductive line **408'** may be formed of the same or substantially the same material (e.g., Ti/Al/Ti) as the source or drain electrode of the thin film transistor. The conductive line **408'** may cover an area including at least one or more grooves in which the inorganic layers are dug. That is, the uneven shape of the conductive line **408'** can be obtained by providing at least one or more grooves by etching or the like, and then, depositing the conductive line **408'** on the groove and a flat portion around the groove. The inorganic layers may include at least one or more of a buffer layer **403**, a gate insulating layer **405**, an interlayer insulating layer **406**, and a protective layer **407**. With reference to FIG. 5, it can be seen that only a portion of the buffer layer **403** is removed when preparing the groove. For example, the buffer layer **403** includes a multi-buffer **403-1** and an active buffer **403-2**, and the active buffer **403-2** is removed from the groove, while the multi-buffer **403-1** may not be removed from the groove, which can be applied as a means for adjusting a depth of the groove. If a deeper groove is needed, even the multi-buffer **403-1** can be removed. The depth, a disposition interval, a width and the like of the groove may vary depending on designing the surface area of the conductive line.

[0080] The organic light emitting display device may further include an organic light emitting element on the planarization layer **409**; and an encapsulation layer **420** covering the organic light emitting element. At this time, the encapsulation layer **420** may include a first inorganic layer **421-1** on the organic light emitting element; an organic layer **422** on the first inorganic layer **421-1**; and a second inorganic layer **421-2** on the organic layer **422**.

[0081] The organic light emitting display device may further include a bank **410** on the planarization layer **409**. The bank **410** divides a boundary between (sub) pixels. That is, an area to which the bank **410** is opened is one light emitting unit as a light emitting area. The bank **410** may also be used to form a blocking structure.

[0082] The organic light emitting display device may further include blocking structures **490-1**, **490-2**, and **490-3** provided to block a flow of the organic layer **422** of the encapsulation layer **420** in the inactive area I/A. A plurality of the blocking structures may be provided. For example, the blocking structures **490-1**, **490-2**, and **490-3** may include a first blocking structure **490-1** which is closest to the active area; a second blocking structure **490-2** which is farther from the active area than the first blocking structure **490-1**; and a third blocking structure **490-3** which is farther from the active area than the second blocking structure **490-2**, that is, which is closer to an end portion of the substrate **401**. In this case, the first blocking structure **490-1** may have a relatively low height, and the second blocking structure **490-2** may have a relatively high height. The first blocking structure **490-1** may be formed of the same or substantially the same material as the planarization layer **409** and the bank **410** of the active area. The second blocking structure **490-2** may be formed by sequentially stacking materials the same as the planarization layer **409**, the bank **410**, and the spacer of the active area. That is, the first blocking structure **490-1** may be stacked in two stages, and the second blocking structure **490-2** may be stacked in three stages. The third blocking structure **490-3** may be formed of the same or substantially the same material as the planarization layer **409** of the active area.

[0083] The conductive line **408'** does not vertically overlap with the blocking structure disposed at the outermost position among the blocking structures. If three blocking structures are disposed as shown in FIG. **4**, the conductive line **408'** does not overlap with the third blocking structure **490-3**. This is because that, as described above with reference to FIG. **3**, moisture permeation due to cracking or delamination may occur in an area where the conductive line and the outermost blocking structure overlap with each other.

[0084] Meanwhile, a metal layer **412'** may be provided to transmit power supplied through the conductive line **408'** to the organic light emitting element. The metal layer **412'** connects the conductive line **408'** and a cathode electrode **416** of the organic light emitting element. At this time, the metal layer **412'** may extend along a side surface (a side wall) of the planarization layer **409**. The metal layer **412'** may be formed of the same or substantially the same material as an anode electrode of the organic light emitting element. The metal layer **412'** may be provided with an outgassing hole for discharging gas generated in the planarization layer under the metal layer **412'**.

[0085] When the conductive line structure of the outer portion is provided as described above, damage to the conductive line due to moisture permeation or cracking can be suppressed. In addition, such a conductive line structure can increase reliability by reducing a driving failure of the display device.

[0086] Although the embodiments of the present disclosure have been described in detail with reference to the accompanying drawings, the present disclosure is not limited thereto and may be embodied in many different forms without departing from the technical concept of the present disclosure. Therefore, the embodiments of the present disclosure are provided for illustrative purposes only but not intended to limit the technical concept of the present disclosure. The scope of the technical concept of the present disclosure is not limited thereto. Therefore, it should be understood that the above-described embodiments are illustrative in all aspects and do not limit the present disclosure. All the technical concepts in the present disclosure and the equivalent scope thereof should be construed as falling within the scope of the present disclosure.

[0087] The various embodiments described above can be combined to provide further embodiments. All of the U.S. patents, U.S. patent application publications, U.S. patent applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and/or listed in the Application Data Sheet are incorporated herein by reference, in

their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments. [0088] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

Claims

1. An organic light emitting display device, comprising: a substrate having an active area and an inactive area adjacent to the active area; an inorganic layer including a multi-buffer layer and an active buffer layer on the substrate; an organic light emitting element in the active area; an encapsulation layer covering the organic light emitting element; at least two or more blocking structures in the inactive area; a touch sensing line and a touch driving line connected to touch electrodes; and a touch insulating layer on the touch sensing line and the touch driving line, wherein the at least two or more blocking structures are configured to block a flow of organic layer of the encapsulation layer, wherein the touch driving line including a plurality of first touch electrodes and first bridges that electrically connect the first touch electrodes, and the touch sensing line including a plurality of second touch electrodes and second bridges that electrically connect the second touch electrodes.
2. The organic light emitting display device of claim 1, wherein the plurality of first touch electrodes are spaced apart at regular intervals along a first direction and the plurality of second touch electrodes are spaced apart at regular intervals along a second direction.
3. The organic light emitting display device of claim 1, wherein the first bridges are disposed on the same layer as the first touch electrodes and electrically connected to the first touch electrode without a separate contact hole.
4. The organic light emitting display device of claim 1, wherein the second bridges are disposed on a layer different from the second touch electrode and electrically connected to the second touch electrode through a contact hole.
5. The organic light emitting display device of claim 1, wherein the touch insulating layer vertically overlaps with a blocking structure disposed at the blocking structures.
6. The organic light emitting display device of claim 1, further comprising a conductive line disposed to have a vertically uneven shape in the inactive area.
7. The organic light emitting display device of claim 6, wherein the conductive line is configured to transmit power to the organic light emitting element.
8. The organic light emitting display device of claim 6, wherein the conductive line at least partially covers an area including at least one groove in the inorganic layer.
9. The organic light emitting display device of claim 1, wherein the inorganic layer further includes at least one of a gate insulating layer, an interlayer insulating layer, and a protective layer.
10. The organic light emitting display device of claim 6, wherein the encapsulation layer includes: a first inorganic layer on the organic light emitting element; an organic layer on the first inorganic layer; and a second inorganic layer on the organic layer.
11. The organic light emitting display device of claim 10, wherein another portion of the encapsulation layer is disposed on the inactive area.
12. The organic light emitting display device of claim 11, wherein the first inorganic layer is disposed on the groove.
13. The organic light emitting display device of claim 10, wherein the conductive line does not vertically overlap with a blocking structure disposed at an outermost position among the blocking structures.

- 14.** The organic light emitting display device of claim 10, wherein the blocking structures include a first blocking structure, a second blocking structure which is farther from the active area than the first blocking structure, and a third blocking structure which is farther from the active area than the second blocking structure, wherein the conductive line does not vertically overlap with the third blocking structure.
- 15.** The organic light emitting display device of claim 14, wherein the first blocking structure is formed of a same or substantially the same material as a planarization layer and a bank of the active area.
- 16.** The organic light emitting display device of claim 14, wherein the second blocking structure is formed by sequentially stacking materials the same as those of a planarization layer, a bank, and a spacer of the active area.
- 17.** The organic light emitting display device of claim 7, wherein the conductive line at least partially fills at least one groove in an inorganic layer.
- 18.** An organic light emitting display device, comprising: a substrate having an active area and an inactive area adjacent to the active area; an inorganic layer including a multi-buffer layer and an active buffer layer on the substrate; an organic light emitting element in the active area; an encapsulation layer covering the organic light emitting element; at least two or more blocking structures in the inactive area; a touch sensing line and a touch driving line connected to touch electrodes; and a touch insulating layer on the touch sensing line and the touch driving line, wherein the at least two or more blocking structures are configured to block a flow of organic layer of the encapsulation layer, wherein the touch driving line including a plurality of first touch electrodes and first bridges that electrically connect the first touch electrodes, and the touch sensing line including a plurality of second touch electrodes and second bridges that electrically connect the second touch electrodes.
- 19.** The organic light emitting display device of claim 18, wherein the plurality of first touch electrodes are spaced apart at regular intervals along a first direction and the plurality of second touch electrodes are spaced apart at regular intervals along a second direction.
- 20.** The organic light emitting display device of claim 18, wherein the first bridges are disposed on the same layer as the first touch electrodes and electrically connected to the first touch electrode without a separate contact hole.
- 21.** The organic light emitting display device of claim 18, wherein the second bridges are disposed on a layer different from the second touch electrode and electrically connected to the second touch electrode through a contact hole.
- 22.** The organic light emitting display device of claim 18, wherein the touch insulating layer vertically overlaps with a blocking structure disposed at an outermost position among the blocking structures.
- 23.** The organic light emitting display device of claim 18, further comprising a conductive line having a vertically uneven shape in the inactive area.
- 24.** The organic light emitting display device of claim 23, wherein the conductive line is configured to transmit power to the organic light emitting element.
- 25.** The organic light emitting display device of claim 23, wherein the conductive line at least partially covers an area including at least one groove in the inorganic layer.
- 26.** The organic light emitting display device of claim 1, further comprising a thin film transistor in the inactive area comprising a gate electrode, source and drain electrodes, and the semiconductor layer, and wherein the active buffer layer is disposed between the multi-buffer layer and a semiconductor layer.
- 27.** The organic light emitting display device of claim 26, wherein the active buffer layer directly contacts the multi-buffer layer and the semiconductor layer.
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